

stage 1:  $F_0 = \log \frac{\# \text{ 1s}}{\# \text{ 0s}}$   
(log-odds)

probability:  $p = \sigma(F) = \frac{1}{1 + e^{-F}}$

pseudo-residuals:  $r = y - p$

update the log-odds:  
 $F \leftarrow F + \eta \cdot \gamma$

leaf value in log-odds:  
 $\gamma = \frac{\sum r}{\sum p(1 - p)}$

regression tree on  $r$

next stage